

Topics in Statistics (III): Statistical Process Control (SPC)
Chapter 6: Univariate Control Charts by Change-Point Detection
(CPD)

Sections 6.1–6.4

Prof. Hong-Ji Yang

NDHU — Dept. of Applied Mathematics (Statistics Division)

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Main idea: detect a shift and estimate where it occurred.

Learning Goals

After this chapter, students should be able to:

- 1 Explain the connection between SPC and change-point detection.
- 2 Estimate a single change-point using likelihood-based criteria.
- 3 Use the CPD mean chart based on $T_{\max,n}$.
- 4 Use the CPD variance chart based on $B_{\max,n}$.
- 5 Use the joint CPD chart based on $J_{\max,n}$.
- 6 Interpret the estimated shift location after a signal.
- 7 Nonparametric RS/P methodology

Chapter Roadmap

Retrospective CPD

- 1 Single change-point
- 2 Likelihood formulation
- 3 Mean-shift estimator
- 4 Multiple change-points
- 5 Dynamic programming idea
- 6 Nonparametric RS/P methodology

Phase II CPD charts

- 1 Mean CPD chart
- 2 Variance CPD chart
- 3 Joint mean–variance CPD chart
- 4 Control-limit calibration
- 5 Post-signal diagnosis

This Section

Motivation and Basic CPD Formulation

Why Change-Point Detection?

SPC objective

Detect a **distributional shift** as soon as possible:

$$F_0 \longrightarrow F_1.$$

The **shift time** is usually unknown:

$$\tau = r + 1.$$

Comparison

Chart	Detect	Estimate
Shewhart	large	rough
CUSUM	small	separate CPD
EWMA	small	separate CPD
CPD	small	built-in

CPD advantage

A CPD chart can

- signal the shift
- simultaneously estimate $\hat{r}$ (**change location**)

Detection + *Localization*

Single Change-Point Model

Mean-shift representation

$$X_i = \begin{cases} \theta_0 + \varepsilon_i, & i = 1, \dots, r, \\ \theta_1 + \varepsilon_i, & i = r + 1, \dots, n, \end{cases}$$

where r is the **change-point**.

Retrospective CPD

n is **fixed**.

Estimate r **after all data are observed**.

Phase II SPC

n increases **sequentially**.

Update the chart **whenever a new X_n arrives**.

Likelihood-Based Change-Point Estimation

Log-likelihood

$$\ell(r, \theta_0, \theta_1) = \sum_{i=1}^r \log f(X_i; \theta_0) + \sum_{i=r+1}^n \log f(X_i; \theta_1).$$

Known parameters

$$\hat{r} = \arg \max_{1 \leq r \leq n-1} \ell(r, \theta_0, \theta_1).$$

Unknown parameters

For **each candidate** r , estimate the before-shift and after-shift parameters, then **choose the best split**.

Normal Mean Shift: Unknown Means

For

$$X_1, \dots, X_r \sim \mathcal{N}(\mu_0, \sigma^2), \quad X_{r+1}, \dots, X_n \sim \mathcal{N}(\mu_1, \sigma^2),$$

with **unknown** μ_0, μ_1 , the change-point estimator is

MLE of the change-point

$$\hat{r} = \arg \max_{1 \leq r \leq n-1} \frac{r(n-r)}{n} (\bar{X}_r - \bar{X}'_r)^2$$

Teaching interpretation

Choose the split that makes the two segment means **most different**, after **accounting for segment sizes**.

Normal Mean Shift: Unknown Means

Assume $X_1, \dots, X_r \sim \mathcal{N}(\mu_0, \sigma^2)$ and $X_{r+1}, \dots, X_n \sim \mathcal{N}(\mu_1, \sigma^2)$ with common variance σ^2 .

Segment mean estimates

For a fixed split r : $\hat{\mu}_0 = \bar{X}_r = \frac{1}{r} \sum_{i=1}^r X_i$, $\hat{\mu}_1 = \bar{X}'_r = \frac{1}{n-r} \sum_{i=r+1}^n X_i$

Change-point estimator

Substituting the MLEs into the likelihood gives

$$\hat{r} = \arg \max_{1 \leq r \leq n-1} \underbrace{\frac{r(n-r)}{n}}_{\text{balance}} \underbrace{(\bar{X}_r - \bar{X}'_r)^2}_{\text{separation}}$$

Why This Formula Works

Mean difference term

$(\bar{X}_r - \bar{X}_r')^2$ measures how far apart the two segment means are. A **larger value** $\Rightarrow$ stronger evidence of a shift.

Size adjustment

$\frac{r(n-r)}{n}$ penalizes highly unbalanced splits; $r \approx \frac{n}{2}$ gives the largest weight.

Interpretation

Choose the split that makes the **two segment means** as different as possible, while ensuring **both segments contain enough data**.

Deriving the Change-point Estimator (I)

Step 1: Model assumption

Assume a **mean shift** at time r :

$$X_1, \dots, X_r \sim N(\mu_0, \sigma^2)$$

$$X_{r+1}, \dots, X_n \sim N(\mu_1, \sigma^2)$$

with

$$\mu_0 \neq \mu_1$$

and common variance σ^2 .

The **likelihood function** is

$$L(r, \mu_0, \mu_1) \propto \exp \left\{ -\frac{1}{2\sigma^2} \left[\sum_{i=1}^r (X_i - \mu_0)^2 + \sum_{i=r+1}^n (X_i - \mu_1)^2 \right] \right\}$$

Step 2: Estimate segment means

For fixed r ,

$$\hat{\mu}_0 = \bar{X}_r$$

$$\hat{\mu}_1 = \bar{X}'_r$$

These are the **MLEs**.

Substitute into the **log-likelihood**:

$$\ell(r) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \text{SSE}(r),$$

$$\text{SSE}(r) = \sum_{i=1}^r (X_i - \bar{X}_r)^2 + \sum_{i=r+1}^n (X_i - \bar{X}'_r)^2$$

Maximize likelihood $\iff$ Minimize SSE

Deriving the Change-point Estimator (II)

Step 3: Expand the SSE

Start with the objective

$$\text{SSE}(r) = \sum_{i=1}^r (X_i - \bar{X}_r)^2 + \sum_{i=r+1}^n (X_i - \bar{X}'_r)^2$$

Use the variance identity

$$\sum (X_i - \bar{X})^2 = \sum X_i^2 - m\bar{X}^2$$

Apply to Segment 1

$$\sum_{i=1}^r (X_i - \bar{X}_r)^2 = \sum_{i=1}^r X_i^2 - r\bar{X}_r^2$$

Apply to Segment 2

$$\sum_{i=r+1}^n (X_i - \bar{X}'_r)^2 = \sum_{i=r+1}^n X_i^2 - (n-r)\bar{X}'_r^2$$

Combine both segments

$$\text{SSE}(r) = \sum_{i=1}^n X_i^2 - r\bar{X}_r^2 - (n-r)\bar{X}'_r^2$$

Since $\sum_{i=1}^n X_i^2$ is **constant with respect to r** ,

$$\min_r \text{SSE}(r) \iff \max_r [r\bar{X}_r^2 + (n-r)\bar{X}'_r^2]$$

Minimize within-group variation



Maximize between-group contrast

Key idea: ignore constants, optimize what changes

Deriving the Change-point Estimator (III)

Step 4: Introduce the Overall Mean

Define the overall mean

$$\bar{X} = \frac{r\bar{X}_r + (n-r)\bar{X}'_r}{n}$$

Rewrite the objective

$$r\bar{X}_r^2 + (n-r)\bar{X}'_r{}^2 = n\bar{X}^2 + r(\bar{X}_r - \bar{X})^2 + (n-r)(\bar{X}'_r - \bar{X})^2$$

Compute the deviations

$$\bar{X}_r - \bar{X} = \frac{n-r}{n}(\bar{X}_r - \bar{X}'_r)$$

$$\bar{X}'_r - \bar{X} = -\frac{r}{n}(\bar{X}_r - \bar{X}'_r)$$

Key insight: both depend on $(\bar{X}_r - \bar{X}'_r)$

Substitute

$$r \left(\frac{n-r}{n} \right)^2 (\bar{X}_r - \bar{X}'_r)^2 \\ + (n-r) \left(\frac{r}{n} \right)^2 (\bar{X}_r - \bar{X}'_r)^2$$

Factor the common term

$$= \frac{r(n-r)}{n} (\bar{X}_r - \bar{X}'_r)^2$$

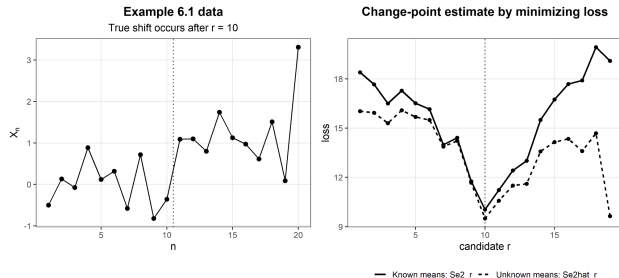
Since $n\bar{X}^2$ is constant,

$$\hat{r} = \arg \max_{1 \leq r \leq n-1} \frac{r(n-r)}{n} (\bar{X}_r - \bar{X}'_r)^2$$

$$\underbrace{\frac{r(n-r)}{n}}_{\text{balance}} \underbrace{(\bar{X}_r - \bar{X}'_r)^2}_{\text{separation}}$$

Best split = balanced groups + large separation

Example 6.1: Single Change-Point



Data

$X_{1:10} \sim N(0, 1), \quad X_{11:20} \sim N(1, 1)$

True shift:

$$r = 10$$

Loss criteria

Known means:

$$Se_r^2 = \sum_{i=1}^r X_i^2 + \sum_{i=r+1}^{20} (X_i - 1)^2$$

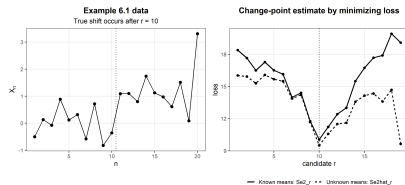
Unknown means:

$$\widehat{Se}_r^2 = \sum_{i=1}^r (X_i - \bar{X}_r)^2 + \sum_{i=r+1}^{20} (X_i - \bar{X}'_r)^2$$

Key result

Both criteria attain their minimum at $\hat{r} = 10$, correctly locating the change-point.

Example 6.1: Scan Statistic for One Change-point



For each candidate split $j = 1, \dots, n - 1$,

$$T_{jn} = \sqrt{\frac{j(n-j)}{n}} \frac{\bar{X}_j - \bar{X}'_j}{\widehat{Se}_j}, \quad \hat{r} = \arg \max_j |T_{jn}|.$$
$$|T_{jn}|^2 = \frac{j(n-j)}{n} \frac{(\bar{X}_j - \bar{X}'_j)^2}{\widehat{Se}_j^2}.$$

Meaning

Large $|T_{jn}|$ means strong between-segment separation relative to within-segment variability.

Example 6.1: Complete Candidate Table

$$\hat{r} = \arg \max_{1 \leq j \leq 19} |T_{jn}|^2 = \arg \min_{1 \leq j \leq 19} \widehat{Se}_j^2$$

$$\text{Known means loss: } Se_{2j} = \sum_{i=1}^j (X_i - \mu_0)^2 + \sum_{i=j+1}^n (X_i - \mu_1)^2$$

$$\text{Estimated loss: } \widehat{Se}_j^2 = \sum_{i=1}^j (X_i - \bar{X}_j)^2 + \sum_{i=j+1}^n (X_i - \bar{X}'_j)^2$$

j	1	2	3	4	5	6	7	8	9	10
$g_j = T_{jn} ^2$	1.297	1.397	2.026	1.242	1.646	1.831	3.452	3.098	5.651	7.830
Estimated	16.028	15.928	15.299	16.083	15.679	15.493	13.873	14.227	11.673	9.495
Known means	18.396	17.660	16.502	17.276	16.510	16.148	13.984	14.414	11.764	10.044

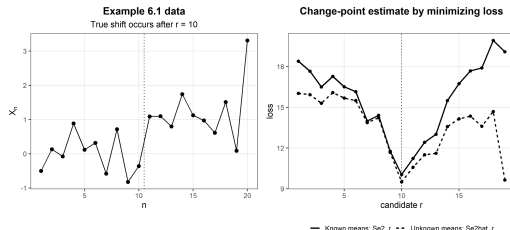
j	11	12	13	14	15	16	17	18	19
$g_j = T_{jn} ^2$	6.737	5.823	5.709	3.742	3.173	2.977	3.728	2.641	7.686
Estimated	10.588	11.502	11.616	13.582	14.152	14.348	13.597	14.684	9.639
Known means	11.224	12.416	13.012	15.492	16.738	17.680	17.902	19.924	19.096

$$\hat{r} = 10$$

Interpretation

Estimated loss uses sample means $(\bar{X}_j, \bar{X}'_j)$, while **Known means loss** uses the true means (μ_0, μ_1) . Both are minimized at $j = 10$, correctly identifying the true change-point.

Example 6.1: Final Interpretation



$$\hat{r} = 10$$

The estimated partition is

$$(1:10), \quad (11:20).$$

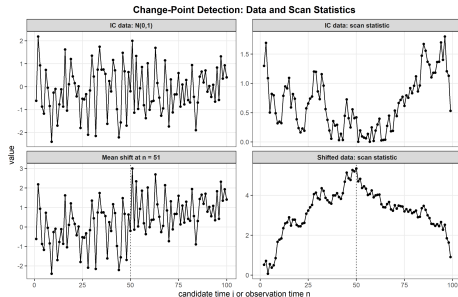
$$\bar{X}_{1:10} = 0.080$$

$$\bar{X}_{11:20} = 1.234$$

Takeaway

The correct split is detected because $j = 10$ gives the strongest mean separation and the smallest within-segment variability.

Interpreting the CPD Scan Statistic



No mean shift (Upper)

$$X_1, \dots, X_{100} \sim N(0, 1)$$

The scan curve has

no clear peak

No evidence of change-point

Mean shift (Lower)

$$X_{1:50} \sim N(0, 1), \quad X_{51:100} \sim N(1, 1)$$

True change-point:

$r = 50$

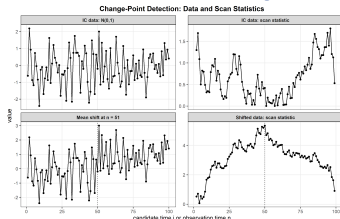
The scan curve shows a sharp peak near $j = 50$

Visual message

Flat curve $\Rightarrow$ no shift

Visible peak $\Rightarrow$ true shift

Why the Scan Statistic Measures Separation



For each candidate split j

$$T_{jn} = \sqrt{\frac{j(n-j)}{n}} \frac{\bar{X}_j - \bar{X}'_j}{\widehat{Se}_j}$$

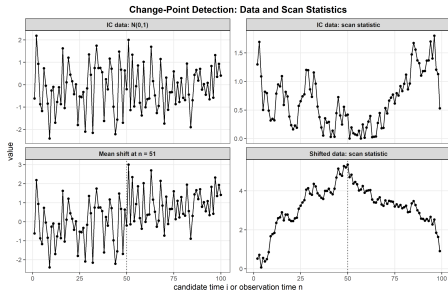
$$\text{where } \widehat{Se}_j^2 = \frac{1}{n-2} \left[\sum_{i=1}^j (X_i - \bar{X}_j)^2 + \sum_{i=j+1}^n (X_i - \bar{X}'_j)^2 \right]$$

$$|T_{jn}| \approx \frac{\text{between-segment difference}}{\text{within-segment variability}}$$

Interpretation

Large $|T_{jn}|$ means **strong separation** at split j .

Estimating the Change-Point



Choose the split with the strongest separation

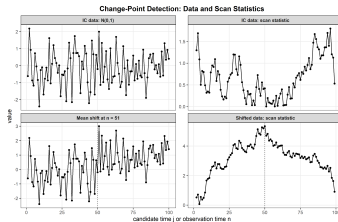
$$\hat{r} = \arg \max_{1 \leq j \leq n-1} |T_{jn}|$$

largest peak $\implies$ estimated change-point

Takeaway

CPD finds the change-point by maximizing the **standardized separation**.

Scan Statistic: Definition



For each candidate split j ,

$$T_{jn} = \sqrt{\frac{j(n-j)}{n}} \frac{\bar{X}_j - \bar{X}'_j}{\widehat{Se}_j}.$$

$$\hat{r} = \arg \max_{1 \leq j \leq n-1} |T_{jn}|.$$

Interpretation

Large $|T_{jn}|$ means strong standardized separation between the two fitted segments.

Complete Scan Statistic Table

$$\hat{r} = \arg \max_{1 \leq j \leq 99} |T_{jn}|$$

j	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
$ T_{jn} $	0.52	0.71	0.07	0.55	0.38	0.49	0.85	1.67	1.78	1.84	2.35	2.59	2.64	2.89	2.48	2.78	2.78	2.51	2.44	2.45	2.58	2.62	3.07	3.23	
j	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
$ T_{jn} $	3.40	3.52	4.05	4.13	3.85	3.80	4.35	4.22	3.87	3.75	3.64	3.58	3.99	4.00	4.10	3.91	3.82	4.10	4.68	5.13	4.85	4.77	5.27	5.19	5.34
j	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
$ T_{jn} $	4.70	4.82	4.37	4.45	4.33	4.01	4.06	4.25	3.89	4.00	4.02	4.04	3.55	3.40	3.39	3.56	3.66	3.30	3.23	3.51	3.33	3.48	3.46	3.43	3.20
j	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	
$ T_{jn} $	3.30	3.18	3.27	3.10	3.15	3.24	2.90	2.93	3.35	3.47	3.27	3.00	2.86	2.57	2.56	2.47	2.55	2.46	2.66	2.27	2.52	1.85	1.63	0.91	

$$\max_j |T_{jn}| = 5.337 \quad \text{at } j = 50$$

Conclusion

The complete scan statistic profile reaches its global maximum at

$$\hat{r} = 50,$$

correctly identifying the true mean shift.

IC Data: Complete Table

$$X_1, \dots, X_{100} \sim N(0, 1)$$

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
X_n	-0.61	2.19	0.93	-0.87	-1.18	0.73	-0.04	-0.84	-2.40	-0.26	-0.10	-1.70	-0.77	-0.11	-0.87	1.62	-1.05	0.10	1.21	0.45	0.16	-0.42	0.02	-1.80	-0.53
n	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
X_n	-0.55	-0.33	-2.09	-0.16	1.35	0.44	-2.15	0.74	1.74	0.74	0.75	0.55	-1.62	0.25	-0.21	1.16	0.72	-0.98	-2.21	-1.56	1.48	0.67	-1.70	0.63	-0.20
n	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
X_n	2.00	-1.13	1.34	-0.97	-0.07	0.86	-0.81	-1.37	1.02	-1.01	-0.64	-0.62	1.69	0.16	-0.48	-1.27	-0.95	1.15	-0.15	-1.73	0.32	-1.12	-0.34	-0.33	0.60
n	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
X_n	-0.87	0.10	-0.77	0.29	-0.58	-0.69	0.94	-0.44	-1.90	-0.69	0.44	0.66	0.19	0.71	-0.22	0.03	-0.45	0.08	-0.65	0.83	-0.58	1.32	0.35	0.93	0.40

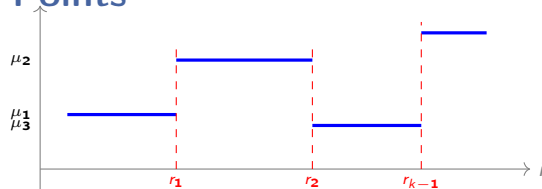
Remark

These are in-control observations generated from $N(0, 1)$; no systematic mean shift is imposed.

This Section

Multiple Change-Points

Multiple Change-Points



Segment model

$$X_i = \begin{cases} \mu_1 + \varepsilon_i, & 1 \leq i \leq r_1, \\ \mu_2 + \varepsilon_i, & r_1 < i \leq r_2, \\ \dots & \\ \mu_k + \varepsilon_i, & r_{k-1} < i \leq n \end{cases}$$

Goal

Find

$$1 \leq r_1 < r_2 < \dots < r_{k-1} \leq n-1$$

to divide the sequence into **homogeneous segments**.

Dynamic Programming (DP): Segment Loss

Step 1: Measure one segment

For observations

$$X_{h+1}, \dots, X_m, \quad 0 \leq h < m$$

define the segment mean

$$\bar{X}_{h,m} = \frac{1}{m-h} \sum_{i=h+1}^m X_i$$

The within-segment loss is

$$Q(h, m) = \sum_{i=h+1}^m (X_i - \bar{X}_{h,m})^2$$

$$Q(h, m) = \text{within-segment variation}$$

Interpretation

A smaller $Q(h, m)$ means the segment is more homogeneous.

Dynamic Programming: Recursive Optimization

Step 2: Build the optimal partition

Let

$$X^2(j, m)$$

denote the minimum total loss for partitioning the first m observations into $1 \leq j \leq m$ segments.

Recursive formula

$$X^2(j, m) = \min_{j-1 \leq h < m} \{X^2(j-1, h) + Q(h, m)\}$$

where $j-1 \leq h < m$ ensures that the first h observations can form $j-1$ nonempty segments.

optimal first $(j-1)$ segments + optimal last segment

Key idea

Dynamic programming solves a large problem by combining optimal smaller subproblems.

Numerical Example: Candidate Split $h = 1$

Consider

$$X_1 = 1, \quad X_2 = 2, \quad X_3 = 8, \quad X_4 = 9$$

$$j = 2, \quad m = 4, \quad h = 1$$

This split gives

$$(X_1) \quad | \quad (X_2, X_3, X_4)$$

First segment

$$\bar{X}_{0,1} = X_1 = 1$$

$$Q(0,1) = \sum_{i=1}^1 (X_i - \bar{X}_{0,1})^2 = (1 - 1)^2 = 0$$

Second segment

$$\bar{X}_{1,4} = \frac{X_2 + X_3 + X_4}{3} = \frac{2 + 8 + 9}{3} = 6.33$$

$$\begin{aligned} Q(1,4) &= \sum_{i=2}^4 (X_i - \bar{X}_{1,4})^2 \\ &= (2 - 6.33)^2 + (8 - 6.33)^2 + (9 - 6.33)^2 \\ &\approx 26.67 \end{aligned}$$

Total loss

$$Q(0,1) + Q(1,4) = 0 + 26.67 = 26.67$$

Numerical Example: Candidate Split $h = 2$

Consider

$$X_1 = 1, \quad X_2 = 2, \quad X_3 = 8, \quad X_4 = 9$$

$$j = 2, \quad m = 4, \quad h = 2$$

This split gives

$$(X_1, X_2) \quad | \quad (X_3, X_4)$$

First segment

$$\bar{X}_{0,2} = \frac{X_1 + X_2}{2} = \frac{1 + 2}{2} = 1.5$$

$$\begin{aligned} Q(0, 2) &= \sum_{i=1}^2 (X_i - \bar{X}_{0,2})^2 \\ &= (1 - 1.5)^2 + (2 - 1.5)^2 \\ &= 0.5 \end{aligned}$$

Second segment

$$\bar{X}_{2,4} = \frac{X_3 + X_4}{2} = \frac{8 + 9}{2} = 8.5$$

$$\begin{aligned} Q(2, 4) &= \sum_{i=3}^4 (X_i - \bar{X}_{2,4})^2 \\ &= (8 - 8.5)^2 + (9 - 8.5)^2 \\ &= 0.5 \end{aligned}$$

Total loss

$$Q(0, 2) + Q(2, 4) = 0.5 + 0.5 = 1.0 \quad (\text{smallest})$$

Numerical Example: Candidate Split $h = 3$

Consider

$$X_1 = 1, \quad X_2 = 2, \quad X_3 = 8, \quad X_4 = 9$$

$$j = 2, \quad m = 4, \quad h = 3$$

This split gives

$$(X_1, X_2, X_3) \quad | \quad (X_4)$$

First segment

Second segment

$$\bar{X}_{0,3} = \frac{X_1 + X_2 + X_3}{3} = \frac{1 + 2 + 8}{3} = 3.67$$

$$\bar{X}_{3,4} = X_4 = 9$$

$$\begin{aligned} Q(0,3) &= \sum_{i=1}^3 (X_i - \bar{X}_{0,3})^2 \\ &= (1 - 3.67)^2 + (2 - 3.67)^2 + (8 - 3.67)^2 \\ &\approx 26.67 \end{aligned}$$

$$Q(3,4) = \sum_{i=4}^4 (X_i - \bar{X}_{3,4})^2 = (9 - 9)^2 = 0$$

Total loss

$$Q(0,3) + Q(3,4) = 26.67 + 0 = 26.67$$

Numerical Example: Why the Best Split is $h = 2$

Step 1: Specify the problem

For

$j = 2$ (number of segments) and

$m = 4$ (observations),

DP searches over

h (candidate split location).

$$X^2(2, 4) = \min_{1 \leq h < 4} \{Q(0, h) + Q(h, 4)\}$$

Step 2: Compare all candidate splits

h	Total loss	Interpretation
1	26.67	poor split
2	1.0	best split
3	26.67	poor split

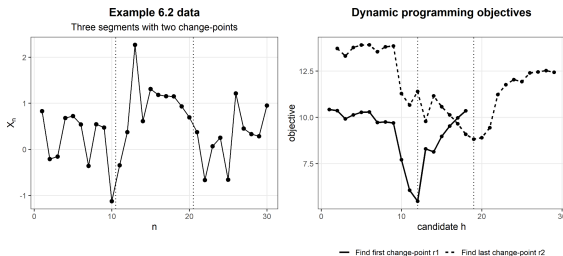
Step 3: Choose the minimum

$$X^2(2, 4) = \min\{26.67, 1.0, 26.67\} = 1.0$$

Takeaway

The minimum occurs at $\hat{h} = 2$ so the optimal partition is $(1, 2) \mid (8, 9)$.

Example 6.2: Dynamic Programming for Two Change-points



Generated data

$$n = 30$$

$$X_{1:10} \sim N(0, 1)$$

$$X_{11:20} \sim N(1, 1)$$

$$X_{21:30} \sim N(0, 1)$$

True structure

$$r_1 = 10, \quad r_2 = 20$$

Three segments: (1:10), (11:20), (21:30)

Mean pattern: $0 \rightarrow 1 \rightarrow 0$

Goal

Use dynamic programming to find the split locations that minimize total within-segment variation.

Example 6.2: Step 1 — Find $\hat{r}_2$

Backward recursion

Since

$$\hat{r}_3 = n = 30,$$

search over all candidate split points

$$\hat{r}_2 = \arg \min_{1 < h < 30} \{X^2(2, h) + Q(h, 30)\}.$$

h	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$X^2(2, h)$	0.000	0.001	0.499	0.684	0.702	1.068	1.331	1.375	1.642	1.950	2.770	3.893	5.269	5.280
$Q(h, 30)$	13.726	13.314	13.279	13.223	13.219	12.483	12.482	12.481	9.646	8.713	8.635	5.897	5.895	5.298
Total	13.726	13.316	13.778	13.907	13.921	13.551	13.813	13.855	11.289	10.663	11.404	9.790	11.163	10.578

h	16	17	18	19	20	21	22	23	24	25	26	27	28	29
$X^2(2, h)$	5.314	5.343	5.363	5.466	5.719	6.276	9.005	9.636	9.938	11.249	12.119	12.179	12.300	12.438
$Q(h, 30)$	4.819	4.309	3.721	3.352	3.182	3.168	2.229	2.130	2.103	0.685	0.282	0.279	0.223	0.000
Total	10.133	9.652	9.084	8.818	8.900	9.443	11.234	11.766	12.041	11.933	12.401	12.458	12.524	12.438

$$\min_{1 < h < 30} \{X^2(2, h) + Q(h, 30)\} = 8.8182$$

$$\hat{r}_2 = 19$$

Stage 1 result

Among all 28 candidate split points, the minimum total objective occurs at

Example 6.2: Step 2 — Find $\hat{r}_1$

Continue backward

After obtaining

$$\hat{r}_2 = 19,$$

search all candidate split points

$$\hat{r}_1 = \arg \min_{0 < h < 19} \{Q(0, h) + Q(h, 19)\}.$$

h	1	2	3	4	5	6	7	8	9
Loss	10.425	10.366	9.924	10.129	10.273	10.289	9.725	9.750	9.701

h	10	11	12	13	14	15	16	17	18
Loss	7.715	6.062	5.466	8.302	8.138	8.976	9.528	9.976	10.353

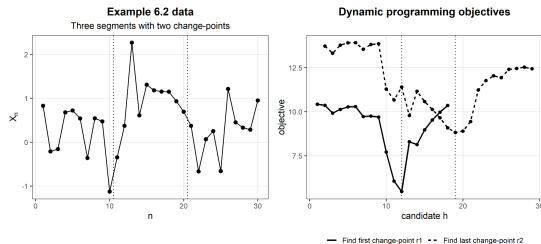
$$\min_{1 \leq h < 19} \{Q(0, h) + Q(h, 19)\} = 5.4659$$

$$\hat{r}_1 = 12$$

Stage 2 result

Among all candidate split points $h = 1, \dots, 18$, the minimum objective occurs at $h = 12$. Thus the first change-point is recovered as $\hat{r}_1 = 12$.

Example 6.2: Final Dynamic Programming Reconstruction



$$\hat{r}_1 = 12, \quad \hat{r}_2 = 19$$

Recovered partition:

(1:12), (13:19), (20:30)

Key algorithmic message

Dynamic programming works by

30 $\rightarrow$ 19 $\rightarrow$ 12

that is, solving backward recursively.

Dynamic Programming Algorithm

Hawkins (2001) proposed a dynamic programming algorithm to estimate

$$\hat{r}_{k-1}, \hat{r}_{k-2}, \dots, \hat{r}_1.$$

Algorithm

Assume that $X_1, \dots, X_n$ satisfy a k -segment change-point model.

Step 1. Compute

$$X^2(j, m), \quad j = 1, \dots, k-1, \quad m = j, j+1, \dots, n-1.$$

Step 2. Set $\hat{r}_k = n$.

For $\ell = 1, \dots, k-1$, find

$$\hat{r}_{k-\ell} = \arg \min_{k-\ell \leq h < \hat{r}_{k-\ell+1}} \{X^2(k-\ell, h) + Q(h, \hat{r}_{k-\ell+1})\}.$$

Interpretation

The algorithm works backward:

$$\hat{r}_k = n \rightarrow \hat{r}_{k-1} \rightarrow \dots \rightarrow \hat{r}_1.$$

Why Add a Distribution-Free Phase I Method?

Phase I objective

Historical **individual observations** are used to check whether the reference sample is **stable**.

stable Phase I data $\Rightarrow$ reliable Phase II limits

Main difficulty

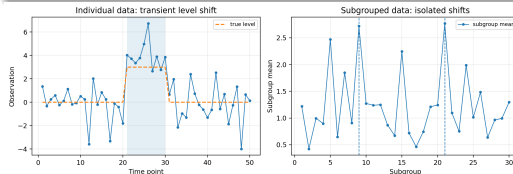
Before stability is established, the **IC distribution** is hard to validate.

Normal-based CPD charts may lose **nominal false-alarm control**.

RS/P solution

recursive segmentation + permutation calibration

Detects **level** and **scale changes** without specifying the IC distribution.



Capizzi and Masarotto (2013), *JQT*

RS/P Phase I Model for Individual Data

Assume ordered Phase I observations:

$$X_1, X_2, \dots, X_m$$

In-control state

$$X_i \sim F_0, \quad i = 1, \dots, m$$

F_0 : unknown

Only **independence** is assumed.

Out-of-control state

$$X_i \sim \begin{cases} F_0, & 1 \leq i \leq \tau_1 \\ F_1, & \tau_1 < i \leq \tau_2 \\ \dots & \\ F_k, & \tau_k < i \leq m \end{cases}$$

Unknown **change-points**

$$\tau_1, \dots, \tau_k$$

$$H_0 : k = 0 \quad \text{vs.} \quad H_1 : k > 0$$

What Does RS/P Try to Detect?

Patterned changes

- step shift
- multiple shifts
- transient change
- gradual drift

$$F_0 \rightarrow F_1 \rightarrow F_2$$

Why not Shewhart alone?

Strong for **isolated outliers**

Less sensitive to **sustained structural changes**

Takeaway

RS/P detects **hidden instability** in Phase I

Phase I Diagnosis: Recursive Segmentation and Permutation

RS/P idea

The RS/P method combines

recursive segmentation + permutation calibration

to detect hidden instability within Phase I data.

Scale statistic

$$T_{\text{scale}}(j) = \sqrt{\frac{j(m-j)}{m}} \left| \log(s_{1:j}^2) - \log(s_{(j+1):m}^2) \right|$$

Large values suggest a possible

variance shift

Level statistic

For each candidate split point j ,

$$T_{\text{level}}(j) = \sqrt{\frac{j(m-j)}{m}} \frac{|\bar{X}_{1:j} - \bar{X}_{(j+1):m}|}{s}$$

Large values suggest a possible

mean shift

Decision rule

Compute

$$\max_j T(j)$$

and compare it with the permutation null distribution.

$$p < 0.05$$

indicates possible internal change.

RS/P: Permutation Calibration

Permutation idea

Randomly shuffle Phase I observations

$$(X_1, \dots, X_m) \rightarrow (X_{\pi(1)}, \dots, X_{\pi(m)})$$

where π is a random reordering.

Under stability

If no change-point exists,

H_0 : stable process

then ordering should not matter.

Shuffle $\Rightarrow$ simulate H_0

Observed statistic

$$T^* = \max_j T(j)$$

Repeat B shuffles

$$T_1^{(perm)}, \dots, T_B^{(perm)}$$

$$B = 1000$$

$$p = \frac{\#\{T_b^{(perm)} \geq T^*\} + 1}{B + 1}$$

$$p < \alpha \Rightarrow \text{change detected}$$

Could random reordering produce a statistic this large?

Toy Example: Permutation Calibration

Observed Phase I data

(4, 5, 6, 7, 20, 21, 22)

Candidate split:

(4, 5, 6, 7) | (20, 21, 22)

$\bar{X}_1 = 5.5, \quad \bar{X}_2 = 21$

$T^* \propto 15.5$

Permutation samples

(21, 5, 7, 4) | (22, 6, 20)

$T_1^{(perm)} \propto 6.75$

(6, 20, 4, 22) | (5, 21, 7)

$T_2^{(perm)} \propto 2$

Question

Is this caused by a

real change

or random ordering?

$$p = \frac{\#\{T_b^{(perm)} \geq T^*\} + 1}{B + 1}$$

few exceed $T^* \Rightarrow$ small p

Permutation compares observed evidence with shuffled no-change patterns.

R Implementation: `dfphase1::rsp()`

RS/P function

The distribution-free Phase I method is implemented by `rsp()` from package `dfphase1`.

Basic usage

```
library(dfphase1)
fit <- rsp(y, L=1000,
           alpha=0.05,
           plot=TRUE)
```

Input

$y = (X_1, \dots, X_m)$ ordered Phase I data

Key tuning

L = permutations, α = significance level

Output

- change-points
- p -values
- diagnostic plot

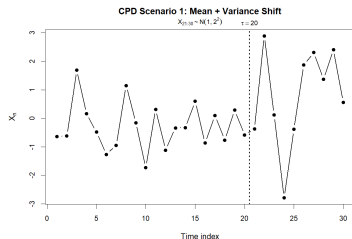
$p < 0.05 \Rightarrow$ possible instability

Source: `dfphase1`

Teaching Setup: Three CPD Scenarios

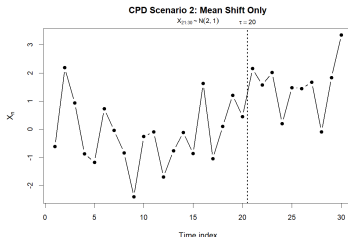
Scenario 1

Mean + variance shift



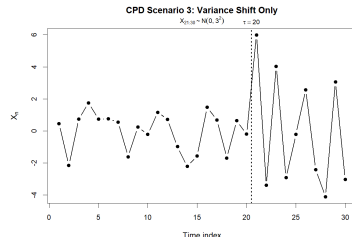
Scenario 2

Mean shift only



Scenario 3

Variance shift only

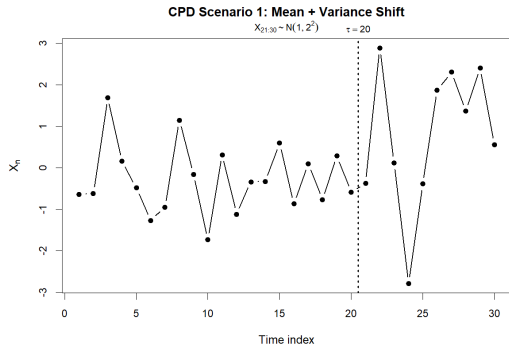


Common structure

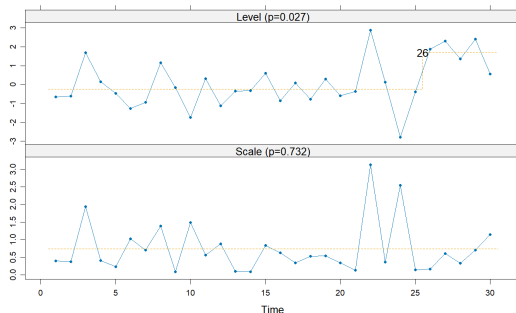
Each sequence has a change point at $\tau = 20$. The purpose is to show how RS/P separates **level evidence** from **scale evidence**.

Scenario 1: Mean + Variance Shift

Raw CPD sequence



RS/P diagnosis

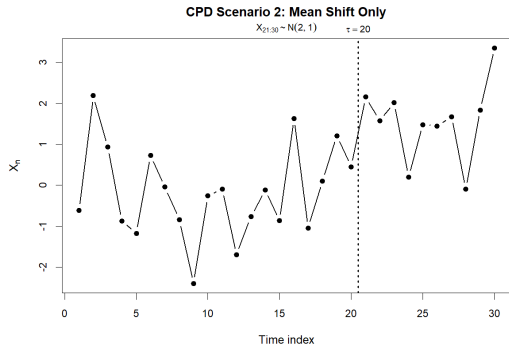


Interpretation

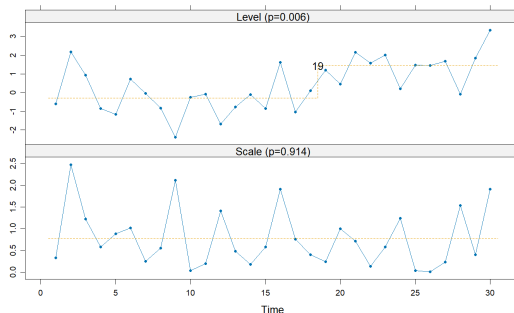
Level test detects a sustained mean displacement: $p_{\text{level}} = 0.027$. Scale evidence is weak here: $p_{\text{scale}} = 0.732$. Thus the RS/P diagnosis is **level-dominant**, even though the data were generated with both mean and variance changes.

Scenario 2: Mean Shift Only

Raw CPD sequence



RS/P diagnosis

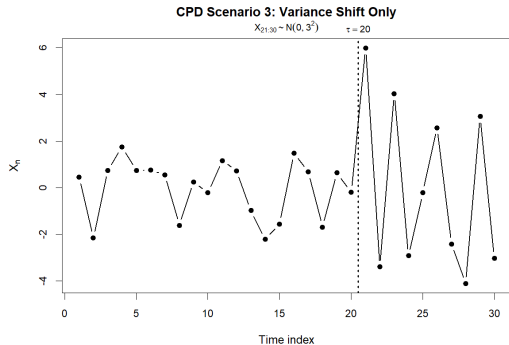


Interpretation

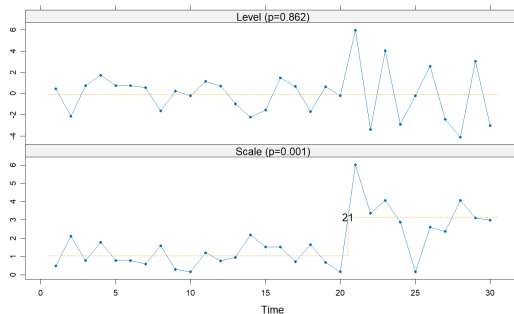
The level panel gives strong evidence of a mean shift: $p_{\text{level}} = 0.006$. The scale panel is not significant: $p_{\text{scale}} = 0.914$. This is the cleanest RS/P pattern for a **pure location shift**.

Scenario 3: Variance Shift Only

Raw CPD sequence



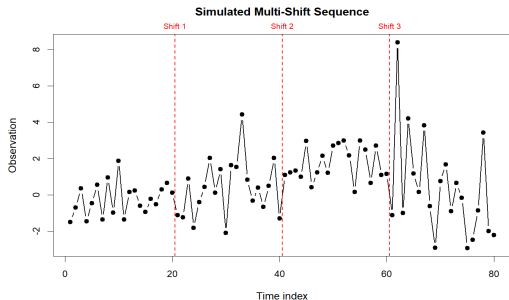
RS/P diagnosis



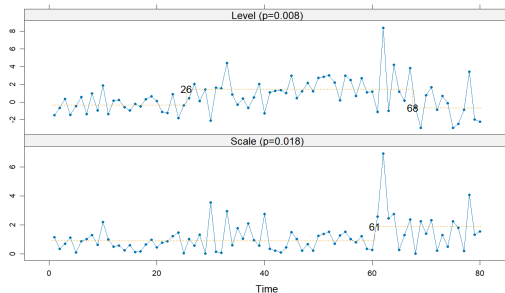
Interpretation

The level panel is not significant: $p_{\text{level}} = 0.862$. The scale panel is highly significant: $p_{\text{scale}} = 0.001$. This provides a clear RS/P diagnosis of a **dispersion/variance shift**.

RS/P Detection of Multiple Shifts



1–20 : $N(0, 1)$
21–40 : $N(1, 2^2)$
41–60 : $N(2, 1)$
61–80 : $N(0, 3^2)$



Level: $p = 0.008$ Scale: $p = 0.018$

Interpretation

- **Level test:** captures shifts from $N(0, 1) \rightarrow N(1, 2^2)$ and $N(1, 2^2) \rightarrow N(2, 1)$
- **Scale test:** detects the volatility jump to $N(0, 3^2)$
- Both $p < 0.05$ indicate significant structural change

RS/P captures both location and scale changes.

This Section

CPD Control Charts for Phase II SPC

From Fixed-Sample CPD to Phase II Monitoring

At time n

Observed data:

$$X_1, \dots, X_n.$$

Test every possible split:

$$j = n_0 - 1, \dots, n - 1.$$

Sequential decision

Compute a maximum statistic:

$$\max_j \{\text{evidence for change at } j\}.$$

Signal if it exceeds h_n .

Mean CPD Chart

Charting statistic (Section 6.3.1)

The charting statistic is the maximum absolute pooled two-sample t -test statistic:

$$T_{\max,n} = \max_{n_0-1 \leq j \leq n-1} |T_{j,n}|$$

where for each split point j :

$$T_{j,n} = \sqrt{\frac{j(n-j)}{n}} \frac{\bar{X}_j - \bar{X}'_j}{S_{j,n}}$$

with sample means $\bar{X}_j = \frac{1}{j} \sum_{i=1}^j X_i$, $\bar{X}'_j = \frac{1}{n-j} \sum_{i=j+1}^n X_i$, and

$$S_{j,n} = \sqrt{\frac{\sum_{i=1}^j (X_i - \bar{X}_j)^2 + \sum_{i=j+1}^n (X_i - \bar{X}'_j)^2}{n-2}}$$

Signal rule

$$T_{\max,n} > h_n$$

After signal ($\hat{r}$)

$$\hat{r} = \arg \max_{n_0-1 \leq j \leq n-1} |T_{j,n}|$$

Recursive Computation for CPD Mean Chart

Motivation

Naive evaluation of $T_{\max,n}$ at every new time point requires **re-computing most quantities**. The computational burden can be **greatly reduced** by using **recursive updates**.

Recursive Updates (6.19)

At time n , define

$$W_n = \sum_{i=1}^n X_i, \quad S_n^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

These can be **updated recursively**:

$$W_n = W_{n-1} + X_n$$

$$S_n^2 = S_{n-1}^2 + \frac{[(n-1)\bar{X}_n - W_{n-1}]^2}{n(n-1)}$$

for $n \geq n_0$.

Efficient Form of the Statistic (6.20)

$$T_{\max,n} = \max_{n_0-1 \leq j \leq n-1} |T_{j,n}|$$

where

$$T_{j,n}^2 = \frac{(n-2)V_{j,n}^2}{S_n^2 - V_{j,n}^2}$$

and

$$V_{j,n}^2 = \frac{(nW_j - jW_n)^2}{nj(n-j)}$$

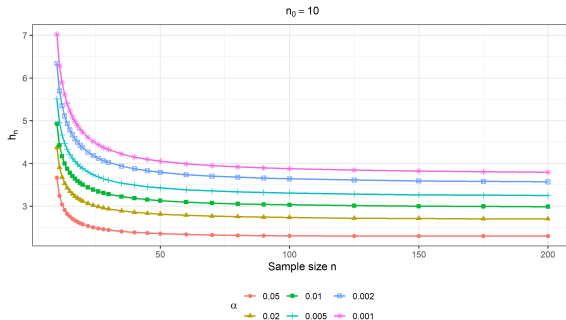
Table 6.2: CPD Mean Chart Control Limits (Hawkins et al., 2003)

CPD mean chart control limits

$n_0 = 10$, $\alpha \in \{0.05, 0.02, 0.01, 0.005, 0.002, 0.001\}$

n	.05	.02	.01	.005	.002	.001
10	3.662	4.371	4.928	5.511	6.340	7.023
11	3.242	3.908	4.424	4.958	5.697	6.284
12	3.037	3.677	4.167	4.664	5.350	5.890
13	2.909	3.530	3.997	4.468	5.110	5.608
14	2.821	3.424	3.875	4.326	4.931	5.397
15	2.756	3.344	3.780	4.211	4.786	5.229
16	2.704	3.281	3.704	4.121	4.671	5.093
17	2.663	3.228	3.642	4.047	4.576	4.977
18	2.628	3.183	3.587	3.981	4.494	4.885
19	2.599	3.146	3.542	3.926	4.425	4.799
20	2.575	3.115	3.503	3.880	4.367	4.730
22	2.535	3.060	3.437	3.800	4.264	4.610
24	2.504	3.019	3.386	3.736	4.187	4.514
26	2.479	2.985	3.343	3.685	4.119	4.440
28	2.459	2.957	3.308	3.643	4.065	4.375
30	2.440	2.933	3.279	3.609	4.024	4.324
40	2.385	2.855	3.184	3.492	3.873	4.147
50	2.355	2.811	3.128	3.426	3.791	4.053
100	2.302	2.735	3.030	3.307	3.640	3.875
150	2.300	2.710	2.997	3.264	3.591	3.821
200	2.300	2.700	2.985	3.248	3.570	3.794

Table 6.2: CPD Mean Chart Control Limits



- h_n decreases rapidly when n is small.
- Smaller α requires larger control limits.
- Curves stabilize as n increases.

Table 6.2: CPD Mean Chart Control Limits (Hawkins et al., 2003)

CPD mean chart control limits

$n_0 = 10, \quad \alpha \in \{0.05, 0.02, 0.01, 0.005, 0.002, 0.001\}$

n	.05	.02	.01	.005	.002	.001
10	3.662	4.371	4.928	5.511	6.340	7.023
11	3.242	3.908	4.424	4.958	5.697	6.284
12	3.037	3.677	4.167	4.664	5.350	5.890
13	2.909	3.530	3.997	4.468	5.110	5.608
14	2.821	3.424	3.875	4.326	4.931	5.397
15	2.756	3.344	3.780	4.211	4.786	5.229
16	2.704	3.281	3.704	4.121	4.671	5.093
17	2.663	3.228	3.642	4.047	4.576	4.977
18	2.628	3.183	3.587	3.981	4.494	4.885
19	2.599	3.146	3.542	3.926	4.425	4.799
20	2.575	3.115	3.503	3.880	4.367	4.730
22	2.535	3.060	3.437	3.800	4.264	4.610
24	2.504	3.019	3.386	3.736	4.187	4.514
26	2.479	2.985	3.343	3.685	4.119	4.440
28	2.459	2.957	3.308	3.643	4.065	4.375
30	2.440	2.933	3.279	3.609	4.024	4.324
40	2.385	2.855	3.184	3.492	3.873	4.147
50	2.355	2.811	3.128	3.426	3.791	4.053
100	2.302	2.735	3.030	3.307	3.640	3.875
150	2.300	2.710	2.997	3.264	3.591	3.821
200	2.300	2.700	2.985	3.248	3.570	3.794

Approximation formula (Hawkins et al., 2003):

$$h_n \approx h_{\alpha}^* \left(0.677 + 0.019 \log(\alpha) + \frac{1 - 0.115 \log(\alpha)}{n - 6} \right)$$

Where: h_{α}^* is a constant depending on α and it equals the computed h_n values when $n = 10$. Namely, it equals 3.662, 4.371, 4.928, 5.511, 6.340, and 7.023, respectively, when α equals 0.05, 0.02, 0.01, 0.005, 0.002, and 0.001.

Example 6.4: Recursive Monitoring Results

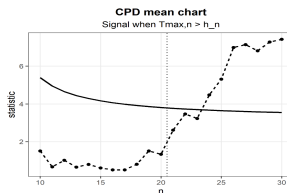
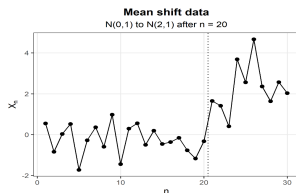
Simulation setting

$$n_0 - 1 = 9, \quad \alpha = 0.005$$

$$X_{1:20} \sim N(0, 1), \quad X_{21:30} \sim N(2, 1)$$

True mean change-point: $\tau = 21$

n	X_n	W_n	S_n^2	$T_{\max,n}$	h_n	Status
10	-1.420	-2.830	4.925	-	4.201	Initialize
11	0.320	-2.510	5.014	1.532	4.105	IC
12	0.580	-1.930	5.226	0.642	4.023	IC
15	-0.370	-1.840	5.813	0.517	3.912	IC
18	-0.760	-2.110	6.245	0.845	3.845	IC
20	-0.280	-2.400	6.513	1.384	3.801	IC
21	1.670	-0.730	8.241	2.634	3.782	Near limit
22	1.410	0.680	9.764	3.482	3.764	Near limit
24	3.720	5.040	14.308	4.521	3.712	Signal
27	2.830	12.740	19.663	6.914	3.621	Signal
30	2.070	18.520	23.017	7.356	3.548	Signal



Monitoring the Process Variance (Section 6.3.2)

Motivation

The previous CPD chart is for detecting **mean shifts only**. However, online monitoring of the **process variance** is equally important for ensuring overall product quality.

Problem Setup

Assume the quality characteristic X follows normal distributions:

- **In-Control (IC):** $X \sim N(\mu_0, \sigma_0^2)$
- **Out-of-Control (OC):** After an unknown time point τ , the distribution shifts to $N(\mu_1, \sigma_1^2)$.

The Challenge

The parameters μ_0, σ_0^2, μ_1 , and σ_1^2 are **all unknown**.

Traditional charts require known IC parameters and struggle when true parameters don't perfectly match the assumptions.

Variance CPD Chart (Hawkins & Zamba)

The CPD Approach

To overcome the limitation of unknown parameters, Hawkins and Zamba (2005a) proposed a **variance control chart** nested within the Change-Point Detection (CPD) framework.

Charting Statistic

Instead of $T_{\max,n}$, the variance chart relies on the statistic $B_{\max,n}$ (derived from Bartlett's test for equal variances), evaluated at each time $n \geq n_0$.

Efficient Computation

Similar to the mean chart, $B_{\max,n}$ is computed efficiently using the recursive updates of:

- $W_n = \sum X_i$
- $S_n^2 = \sum (X_i - \bar{X}_n)^2$

Signal Rule (Eq. 6.25 – 6.26)

A variance shift is detected if: $B_{\max,n} > h_n$, where h_n is a control limit computed via an approximation formula for a given significance level α (e.g., $\alpha = 0.005$).

Variance CPD Chart

Bartlett-type statistic (Bartlett, 1937, 1955)

$$B_{rn} = \frac{1}{C_{rn}} \left[(r-1) \log \left(\frac{\hat{\sigma}^2}{\hat{\sigma}_0^2} \right) + (n-r-1) \log \left(\frac{\hat{\sigma}^2}{\hat{\sigma}_1^2} \right) \right].$$

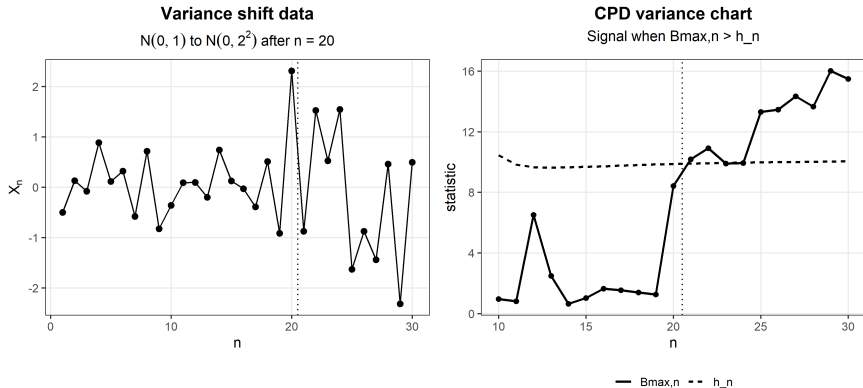
Variance CPD chart

$$B_{\max,n} = \max_{n_0+1 \leq r \leq n-2} B_{rn}$$

Signal when

$$B_{\max,n} > h_n.$$

Example 6.5: CPD Variance Chart



Interpretation

The statistic $B_{\max,n}$ is designed for variance changes, including upward or downward variance shifts.

Derivation of the Bartlett-type Statistic

Step 1: Alternative model

$$X_{1:r} \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad X_{r+1:n} \sim \mathcal{N}(\mu_1, \sigma_1^2)$$

Log-likelihood:

$$\ell_1 = -\frac{r-1}{2} \log(\hat{\sigma}_0^2) - \frac{n-r-1}{2} \log(\hat{\sigma}_1^2) + \text{const}$$

where

$$\hat{\sigma}_0^2 = \frac{Se_r^2}{r-1}, \quad \hat{\sigma}_1^2 = \frac{Se_r'^2}{n-r-1}$$

Step 2: Null model

$$H_0: \sigma_0^2 = \sigma_1^2 = \sigma^2$$

$$\ell_0 = -\frac{n-2}{2} \log(\hat{\sigma}^2) + \text{const}$$

$$\hat{\sigma}^2 = \frac{\widehat{Se}_r^2}{n-2}$$

Step 3: Likelihood ratio

$$-2 \log \Lambda = -2(\ell_0 - \ell_1)$$

$$= (r-1) \log\left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}_1^2}\right) + (n-r-1) \log\left(\frac{\hat{\sigma}_1^2}{\hat{\sigma}_0^2}\right)$$

Step 4: Bartlett correction

$$B_{rn} = \frac{-2 \log \Lambda}{C_{rn}}$$

$$C_{rn} = 1 + \frac{1}{3} \left(\frac{1}{r-1} + \frac{1}{n-r-1} - \frac{1}{n-2} \right)$$

Result

$$B_{rn} \sim \chi_1^2$$

$B_{rn} \uparrow \Rightarrow$ variance change-point

Variance CPD Chart: Control Limits h_n

Table 6.5: $n_0 = 10$

n	.05	.02	.01	.005	.002	.001
10	6.374	8.003	9.229	10.451	12.039	13.238
11	5.651	7.328	8.585	9.840	11.489	12.734
12	5.357	7.077	8.373	9.653	11.357	12.631
13	5.228	6.988	8.312	9.634	11.367	12.672
14	5.173	6.960	8.304	9.658	11.423	12.760
15	5.149	6.960	8.323	9.692	11.469	12.828
16	5.141	6.974	8.357	9.731	11.541	12.885
18	5.142	7.010	8.413	9.808	11.651	13.034
20	5.150	7.034	8.458	9.875	11.722	13.120
24	5.173	7.085	8.529	9.961	11.853	13.297
28	5.196	7.125	8.585	10.035	11.947	13.385
30	5.204	7.136	8.610	10.065	11.981	13.408

Teal: direct-use rows ($n \leq 15$); : approximation range

Control-limit rule

$$P(B_{\max,n} > h_n \mid \text{no prev. signal}) \leq \alpha$$

$$\text{ARL}_0 = \frac{1}{\alpha}$$

Use table values ($n = 10, \dots, 15$)

Read h_n from Table 6.5 directly.

Approximation for $n > 15$

$$h_n \approx \underbrace{-1.38 - 2.241 \log \alpha}_{\text{intercept \& slope}} + \frac{1.61 + 0.691 \log \alpha}{\sqrt{n-9}}$$

$$\alpha \in [0.001, 0.05]$$

Special case $\alpha = 0.05$:

$$h_n \approx 5 + 0.066 \log(n-9)$$

Recursive Computation of B_{rn}

Step 1: Maintain two recursive quantities

$$W_n = \sum_{k=1}^n X_k, \quad S_n^2 = \sum_{k=1}^n (X_k - \bar{X}_n)^2$$

$$W_n = W_{n-1} + X_n$$

$$S_n^2 = S_{n-1}^2 + \frac{[(n-1)X_n - W_{n-1}]^2}{n(n-1)}$$

Key computational idea

Only update W_n and S_n^2 when a new observation X_n arrives.

Step 2: For each candidate split r

$$\bar{X}'_r = \frac{W_n - W_r}{n - r}$$

$$\tilde{S}_r'^2 = S_n^2 - \tilde{S}_r^2 - \frac{r(n-r)}{n}(\bar{X}_r - \bar{X}'_r)^2$$

$$\hat{\sigma}_0^2 = \frac{\tilde{S}_r^2}{r-1}, \quad \hat{\sigma}_1^2 = \frac{\tilde{S}_r'^2}{n-r-1}$$

$$B_{rn} = \frac{(r-1) \log(\hat{\sigma}^2 / \hat{\sigma}_0^2) + (n-r-1) \log(\hat{\sigma}^2 / \hat{\sigma}_1^2)}{C_{rn}}$$

Monitoring statistic

$$B_{\max, n} = \max_{n_0+1 \leq r \leq n-2} B_{rn}$$

Example 6.5: Recursive Monitoring Results

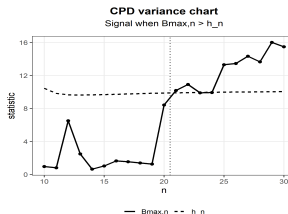
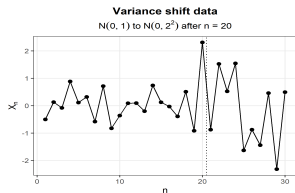
Simulation setting

$$n_0 - 1 = 9, \quad \alpha = 0.005$$

$$X_{1:20} \sim N(0, 1), \quad X_{21:30} \sim N(0, 2^2)$$

True variance change-point: $\tau = 21$

n	X_n	W_n	S_n^2	$B_{\max, n}$	h_n	Status
10	-0.360	-0.180	3.065	—	10.451	Initialize
11	0.090	-0.090	3.075	0.399	9.840	IC
12	0.096	0.007	3.085	6.484	9.653	IC
15	0.123	0.668	3.659	1.186	9.692	IC
18	0.842	1.245	6.824	4.912	9.808	IC
20	2.310	2.157	10.042	8.023	9.875	IC
21	-0.876	1.281	10.964	9.685	9.901	Near limit
22	1.528	2.809	13.018	10.367	9.925	Signal
25	-1.629	3.251	18.274	12.657	9.981	Signal
30	0.494	-0.426	27.354	14.762	10.046	Signal



Joint Mean–Variance CPD Chart: Setup (Part 1)

Step 1: Hypotheses

$H_0: X_i \sim N(\mu_0, \sigma_0^2)$ for all $i = 1, \dots, n$

H_1 : a **change-point** at unknown $r \in \{1, \dots, n-1\}$:

$$X_i \sim \begin{cases} N(\mu_0, \sigma_0^2), & i = 1, \dots, r, \\ N(\mu_1, \sigma_1^2), & i = r+1, \dots, n, \end{cases} \quad \text{where } \mu_0 \neq \mu_1 \text{ and/or } \sigma_0^2 \neq \sigma_1^2.$$

Step 2: Variance Components

$$S_n^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2 \quad \text{Total SS under } H_0$$

$$\tilde{S}_r^2 = \sum_{i=1}^r (X_i - \bar{X}_r)^2 \quad \text{Pre-change SS (first segment)}$$

$$\tilde{S}_r'^2 = \sum_{i=r+1}^n (X_i - \bar{X}_r')^2 \quad \text{Post-change SS (second segment)}$$

Joint Mean–Variance CPD Chart: Setup (Part 2)

Step 3: Likelihood-Ratio Statistic (Bartlett-Corrected)

$$J_{rn} = \frac{1}{C_{rn}^*} \left[r \log \frac{S_n^2/n}{\tilde{S}_r^2/(r-1)} + (n-r) \log \frac{S_n^2/n}{\tilde{S}_{n-r}^2/(n-r-1)} \right]$$

$$C_{rn}^* = 1 + \frac{11}{12} \left(\frac{1}{r} + \frac{1}{n-r} - \frac{1}{n} \right)$$

Step 4: Charting Statistic

$$J_{\max,n} = \max_{n_0+1 \leq r \leq n-2} J_{rn}$$

Range $n_0 + 1 \leq r \leq n - 2$ ensures both segments have sufficient observations for J_{rn} to be well-defined.

Key Contrast: J_{rn} vs. B_{rn}

J_{rn} : sensitive to **mean AND variance** shifts

B_{rn} : sensitive to **variance shifts only**

Summary Joint Mean–Variance CPD Chart: Derivation of J_n

1. Statistical Hypotheses

- H_0 (**In-control**): All $X_1, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$.
- H_1 (**Out-of-control**): A change occurs at unknown time r . The first r observations are $N(\mu_0, \sigma_0^2)$ and the remaining $n - r$ are $N(\mu_1, \sigma_1^2)$.

2. Sum of Squares Components The modified likelihood-ratio test replaces pure MLE variances with unbiased estimators. We define three sums of squares:

- *Total*: $S_n^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2$
- *Pre-change*: $\tilde{S}_r^2 = \sum_{i=1}^r (X_i - \bar{X}_{1:r})^2$
- *Post-change*: $\tilde{S}_r'^2 = \sum_{i=r+1}^n (X_i - \bar{X}_{r+1:n})^2$

3. The Bartlett-Corrected Statistic To improve the χ^2 approximation of the log-likelihood ratio for small samples, we apply a Bartlett correction factor, C_{rn}^* . For a postulated change-point r :

$$J_{rn} = \frac{1}{C_{rn}^*} \left[r \log \frac{S_n^2/n}{\tilde{S}_r^2/(r-1)} + (n-r) \log \frac{S_n^2/n}{\tilde{S}_r'^2/(n-r-1)} \right]$$

$$\text{where } C_{rn}^* = 1 + \frac{11}{12} \left(\frac{1}{r} + \frac{1}{n-r} - \frac{1}{n} \right)$$

4. Final Monitoring Statistic Because the true change-point is unknown, we scan all valid split points (requiring enough observations to estimate variances) and maximize:

Joint Mean–Variance CPD Chart: Signal and Limits (Part 1)

Step 1: Signal Rule

$J_{\max,n} > h_n$ **Control limit** $h_n > 0$ chosen at level α :

$$P(J_{\max,n} > h_n \mid \text{no signal before } n) \leq \alpha, \quad \text{ARL}_0 = \frac{1}{\alpha}.$$

Step 2: Tabulated Control Limits h_n for $n_0 = 9$ (Table 6.7)

n	.05	.02	.01	.005	.002	.001
10	10.128	12.237	13.795	15.330	17.352	18.840
11	9.213	11.389	12.996	14.556	16.609	18.173
12	8.854	11.083	12.719	14.313	16.397	17.965
13	8.690	10.961	12.631	14.265	16.353	17.950
14	8.616	10.917	12.610	14.249	16.361	17.978

Step 3: Approximation for $n > 14$

$$h_n \approx \begin{cases} 1.58 - 2.52 \log \alpha + \frac{0.094 + 0.33 \log \alpha}{\sqrt{n-9}}, & \alpha \in [0.001, 0.05), \\ 8.43 + 0.074 \log(n-9), & \alpha = 0.05. \end{cases}$$

Joint Mean–Variance CPD Chart: Signal and Limits (Part 2)

Step 4: Recursive Computation

Maintain the sequences $\{W_n\}$ and $\{S_n^2\}$ updated **recursively** via (6.19) as each new observation X_n arrives. This is the **same structure** as the $B_{\max,n}$ chart — there is no need to recompute from scratch at each step.

$$W_n = W_{n-1} + X_n,$$

$$S_n^2 = S_{n-1}^2 + \frac{\{(n-1)X_n - W_{n-1}\}^2}{n(n-1)}.$$

Step 5: Estimated Change-Point After Signal

Once $J_{\max,n} > h_n$ signals at time n , the **estimated change-point** is the maximiser:

$$\hat{r} = \arg \max_{n_0+1 \leq r \leq n-2} J_{rn}.$$

$\hat{r}$ identifies which observation most likely marks the beginning of the shift.

Post-Signal Root-Cause Analysis (Part 1)

Context

After a signal at time n with estimated change-point $\hat{r}$: diagnose whether the shift is in the **mean**, the **variance**, or **both**.

Step 1: Variance-Shift Test — *F*-Statistic

$$F = \frac{\tilde{S}_r^2 / (\hat{r} - 1)}{\tilde{S}'_r{}^2 / (n - \hat{r} - 1)} \sim F_{\hat{r}-1, n-\hat{r}-1}.$$

Large F : variance *increase*

Small F : variance *decrease*

Step 2: Mean-Shift Test — *Welch t*-Statistic

$$t = \frac{\bar{X}_{\hat{r}} - \bar{X}'_{\hat{r}}}{\sqrt{\hat{\sigma}_0^2 / \hat{r} + \hat{\sigma}_1^2 / (n - \hat{r})}} \sim t_k$$

where the Welch degrees of freedom are: $k = \frac{(\hat{\sigma}_0^2 / \hat{r} + \hat{\sigma}_1^2 / (n - \hat{r}))^2}{\frac{(\hat{\sigma}_0^2 / \hat{r})^2}{\hat{r} - 1} + \frac{(\hat{\sigma}_1^2 / (n - \hat{r}))^2}{n - \hat{r} - 1}}.$

Post-Signal Root-Cause Analysis (Part 2)

Step 3: Caution — Reliability of Post-Signal Tests

These null distributions assume the **true change-point r is known**. With the estimated $\hat{r}$, three issues arise:

- (a) **Distributional shift:** Null distributions may **differ substantially**.
- (b) **Short post-signal segment:** $n - \hat{r}$ is typically **small**.
- (c) **High variability:** Both F and t may be **highly unstable**.

Step 4: Diagnosis Summary

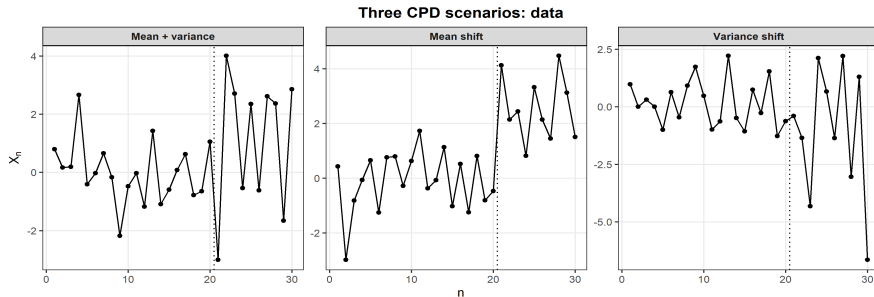
Large F , large $|t|$: both mean and variance shifted

Large F , small $|t|$: variance shift only

Small F , large $|t|$: mean shift only

Use F and t as **informal references only**, not as formal inferential tests.

Simulation Setup: Three CPD Scenarios



Baseline Setup & Shift Scenarios

- **Sequence:** $n = 30$ observations. The first **20 observations** are **in-control**: $X_1, \dots, X_{20} \sim N(0, 1)$.
Change-point: $\tau = 20$. The **shift starts** from observation 21, marked by the **dotted line**.
- **Three post-change scenarios:**
 - 1 **Mean + Variance shift:** $X_{21}, \dots, X_{30} \sim N(1, 2^2)$.
 - 2 **Mean shift only:** $X_{21}, \dots, X_{30} \sim N(2, 1)$.
 - 3 **Variance shift only:** $X_{21}, \dots, X_{30} \sim N(0, 3^2)$.

Joint Monitoring: Chart Performance Summary

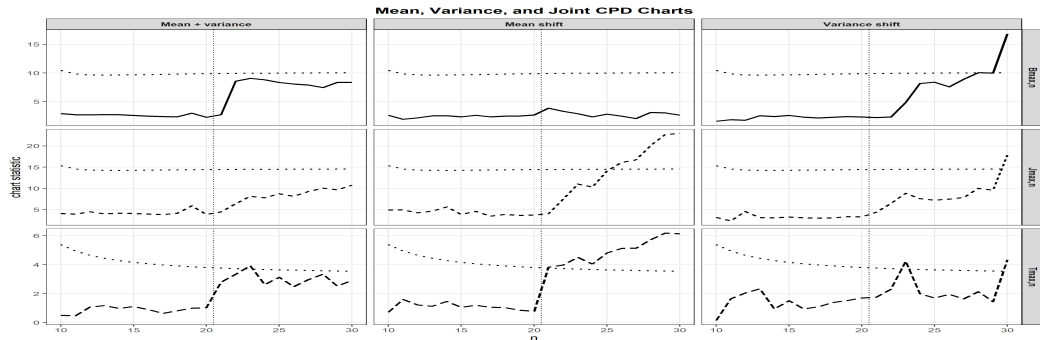


Chart	Mean + Variance (Left)	Mean Shift Only (Middle)	Variance Shift Only (Right)
Variance Chart ($B_{\max,n}$)	No Signal	No Signal	Signal
Joint Chart ($J_{\max,n}$)	No Signal	Strong Signal	Signal
Mean Chart ($T_{\max,n}$)	Signal	Signal	Weak Signal

Teaching Note: The Joint Chart ($J_{\max,n}$) detects isolated shifts strongly. In the left scenario, the combined shift magnitude was too small to breach the joint control limit individually.

Key Formulas to Remember

Single mean change-point

$$\hat{r} = \arg \max_{1 \leq r \leq n-1} \frac{r(n-r)}{n} (\bar{X}_r - \bar{X}'_r)^2.$$

Phase II mean CPD chart

$$T_{\max,n} = \max_j |T_{j,n}|, \quad \text{signal if } T_{\max,n} > h_n.$$

Variance and joint charts

$$B_{\max,n} = \max_r B_{rn}, \quad J_{\max,n} = \max_r J_{rn}.$$

Summary: CPD Charts

The Core Advantage

CPD charts not only **detect** a shift, but also **estimate its exact location** (the unknown change-point r).

- **Sequential Operation:** Phase II CPD charts update statistics sequentially as new data arrives.
- **Targeted Test Statistics:**
 - $T_{\max,n}$ targets **mean** shifts.
 - $B_{\max,n}$ targets **variance** shifts.
 - $J_{\max,n}$ targets **both** (mean and/or variance).
- **Actionable Insights:** Provides rich **post-signal diagnostic** information for root-cause analysis.

CPD vs Conventional Control Charts

Feature	Shewhart	CUSUM / EWMA	CPD
Primary Goal	Detect large abrupt shifts	Detect small persistent shifts	Detect and locate change-points
Historical Information	Current observation only	Weighted recent history	Entire observed sequence
Design Parameters	Control limits	Reference / smoothing constants	Threshold h_n
Shift Localization	Not explicit	Requires post-signal analysis	Directly estimated
Computational Cost	Low	Moderate	Higher
Best Use Case	Sudden large shifts	Gradual small drifts	Unknown structural

This Section

Final Exam Project

Final Exam Project: Dataset and Objective

Project dataset

Use the **UCI SECOM semiconductor manufacturing dataset**

<https://www.kaggle.com/datasets/paresh2047/uci-semcom?resource=download>

1567 observations 591 process variables

Each student selects **one usable variable**.

Project objective

Construct a complete SPC workflow

Phase I diagnosis → Control chart design → Phase II monitoring

Main task

Compare the detection performance of multiple SPC methods on real semiconductor process data.

Phase I / Phase II Design and Required Methods

Phase split

Let

$$X_1, \dots, X_n$$

Define

$$m = \lfloor 0.5n \rfloor$$

$$\underbrace{X_1, \dots, X_m}_{\text{Phase I}} \quad | \quad \underbrace{X_{m+1}, \dots, X_n}_{\text{Phase II}}$$

Phase I diagnosis

Remove missing values and outliers, then apply the **RS/P recursive segmentation + permutation method** Report:

$$p_{\text{level}}, \quad p_{\text{scale}}$$

Phase II monitoring

- Individuals Shewhart (benchmark)
- mean-EWMA, Variance-EWMA, Adaptive-EWMA
- mean-CUSUM, Variance-CUSUM
- Sequential CPD: mean CPD chart, variance and joint charts

Compare under: mean-shift, variance-shift

Deliverables and Final Submission

Required report contents

- 1 Variable summary
- 2 RS/P diagnosis
- 3 Monitoring plots
- 4 Signal comparison table
- 5 Interpretation
- 6 Conclusion

Files to submit

- One **text report**
- One **PPT report**

Submission deadline

June 16, 2026 12:00

Final analysis question

Determine:

Which chart signals first?

Interpret:

mean shift / variance shift / structural change

Final Exam Project Presentation

Phase I/II Statistical Process Monitoring

Final Project Report Presentation

Presentation Schedule

June 16, 2026 13:10

Presentation Format

- 20-minute presentation
- 10-minute Q&A
- Individual report

Participation Rule

Each student must ask

at least one question

to every presenter.

Evaluation Focus

Technical correctness Interpretation Presentation clarity Q&A participation

Final Reminder

Before submission

Please check that your final files clearly include:

- selected SECOM variable and preprocessing rule;
- Phase I RS/P level and scale diagnosis;
- Phase II monitoring results from Shewhart, EWMA, CUSUM, and CPD;
- comparison of first signal time and signal frequency;
- concise interpretation of whether the pattern suggests a mean shift, variance shift, or structural change.

Key deadlines

Submit: June 16, 2026, 12:00

Present: June 16, 2026, 13:10